COMPSYS 306 Project 2 Group 20 – Week 10: Preprocessing and Hyperparameter Tuning

1. Image Preprocessing

The dataset contains six classes: blue light, road, sheep, speed, stop and yellow light. Images were resized to 64 × 48 pixels, converted to RGB format and normalised by dividing pixel values by 255, producing values between 0 and 1. Each image was then flattened into a 9,216-dimensional feature vector for classification. The dataset was divided into training, validation and test sets. These preprocessing steps ensured that all images had a consistent size and format, allowing them to be used as input to the Support Vector Machine (SVM).

2. SVM Hyperparameter Tuning

An SVM with a radial basis function (RBF) kernel was used for image classification. Hyperparameter tuning evaluated 16 combinations of the regularisation parameter C and kernel coefficient gamma. The tested values of C were 0.1, 1, 10 and 100, while the tested values of gamma were scale, 0.001, 0.01 and 0.1.

The best-performing configuration was C=1 and gamma=0.001, achieving a validation accuracy of approximately 95.14%. Configurations using gamma=0.1 performed considerably worse, achieving only 46.50% validation accuracy across the tested values of C. A large gamma narrows the influence of each support vector, so the model overfits the training images. The identical accuracy for every C suggests it collapsed to predicting a single class. Thus demonstrating that the choice of hyperparameters significantly affects classification performance. In particular, an appropriate gamma value is important for controlling the complexity of the nonlinear decision boundary.

3. Selected Model and Conclusion

The selected model was an RBF-kernel SVM with C = 1 and gamma=0.001. It achieved approximately 95.14% validation accuracy and 95.17% test accuracy. On the test set of 331 images, the model correctly classified approximately 95% of the traffic-sign images. Its macro-averaged precision was approximately 96%, macro-averaged recall was 93%, and macro-averaged F1-score was 95%. These results indicate strong overall classification performance across the six classes.

The yellow light recognition remained more challenging, with a test recall of 82%. One possible explanation is the variation in camera conditions during operation. The training dataset includes both still-camera and motion-camera images to account for changes in brightness, exposure and image quality. These variations are important because the JetBot may wait at a blue light before continuing to move when the signal changes to yellow. Differences in camera exposure and motion conditions may make yellow light images more difficult to classify consistently. Including both capture conditions aims to improve the model's robustness in real-world operation, although further testing is required to determine their effect on classification performance.